Shell exercise

Task Description

Students must create a smart backup utility. The script accepts three parameters:

- A source directory S
- A target backup directory T
- A time limit of D days

[nosep]

The script must:

- Traverse the source
- Identify files modified within the last D days
- Copy them to the backup location.

To save disk space, the script must implement primitive data deduplication: if a file with the exact same content (verified by a cryptographic hash) already exists anywhere in the backup archive, create a hard link instead of copying the file again. Cryptographic hashes must be generated using the command `md5sum`.

The script must be validated by the provided `validator.sh` script. The validator and its support data files must be uncompressed in a directory. The validator must be called from that directory with the name of the script as the only parameter.

Steps

- Create a function that extracts the list of files to process (for testing purposes the list must be also saved on a file named `files.txt`). Unreadable files, symlinks or inaccessible directories must be silently skipped.
- Create a function that, given a list of filenames through the standard input, returns on the standard output the file name and, if it exists, the first file with the same content. The second field is separated by "`:`". For testing purposes, the output must be saved on a file named `duplicates.txt`. The output format should match the following logic:

Input	**Output**
F1	F1
F2	F2:F1
F3	F3
F4	F4:F3

- Create a function that, given the output of the previously defined function, copies or links the files.
- Test the functions with a basic environment.
- Test the function with symlinks, unreadable files, inaccessible directories, and files/directories having spaces in their names.